PHYS205 Homework 5

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Question 1

Consider an electromagnetic wave composed of two orthogonal electric field components:

$$E_x(t) = E_0 cos(\omega t) \quad E_y(t) = E_0 cos(\omega t + \Delta\phi)$$

where $\Delta\phi$ is the phase difference between the components.

a) Write the parametric equations describing the trajectory of the electric-field vector

$$\vec{E}(t) = (E_x(t),\ E_y(t))$$

in the xy-plane.

Solution:

Answer

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 cos(\omega t + \Delta\phi)\hat{j}$$

b) For the following phase differences:

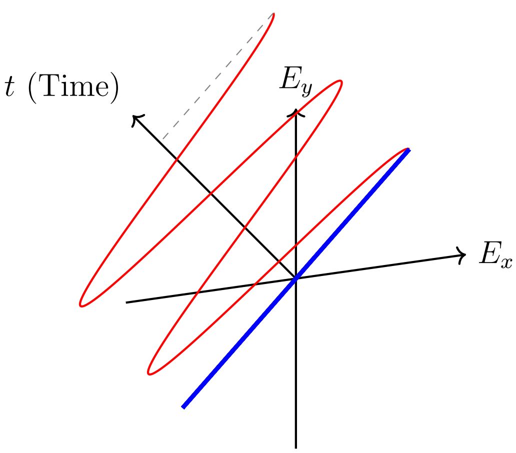
$$\Delta\phi = 0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{5\pi}{6}, \pi$$

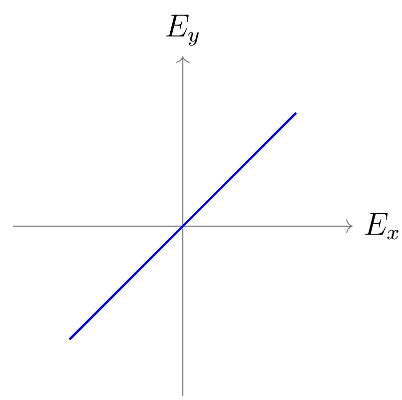
plot the polarization trajectories $(E_x(t),\ E_y(t))$. Your plots should illustrate the progression from **linear** to **elliptical** to **circular** polarization.

Solution:

For $\Delta\phi = 0$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 cos(\omega t)\hat{j}$$
$$\vec{E}(t) = E_0 cos(\omega t) \cdot (\hat{i} + \hat{j})$$

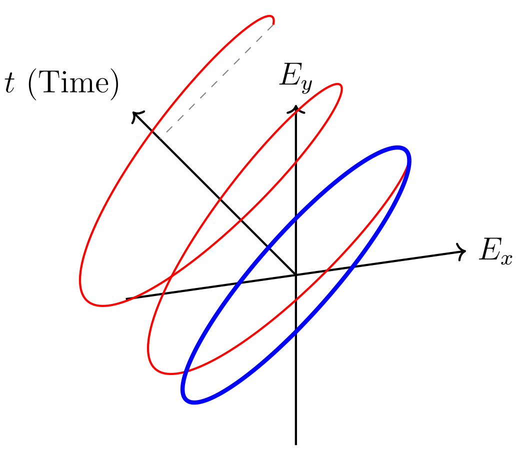


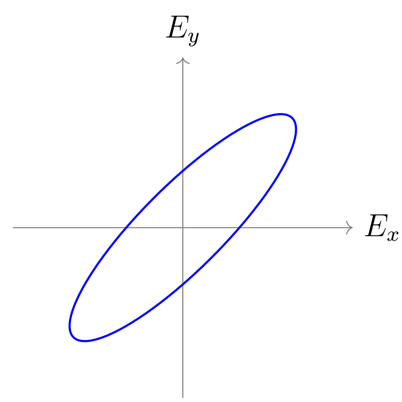


Linear ($\Delta\phi = 0$)

For $\Delta\phi = \frac{\pi}{6}$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 cos(\omega t + \frac{\pi}{6})\hat{j}$$
$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + [E_0 cos(\omega t) cos(\frac{\pi}{6}) - E_0 sin(\omega t) sin(\frac{\pi}{6})]\hat{j}$$
$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + [E_0 cos(\omega t) \cdot \frac{\sqrt{3}}{2} - E_0 sin(\omega t) \cdot \frac{1}{2}]\hat{j}$$





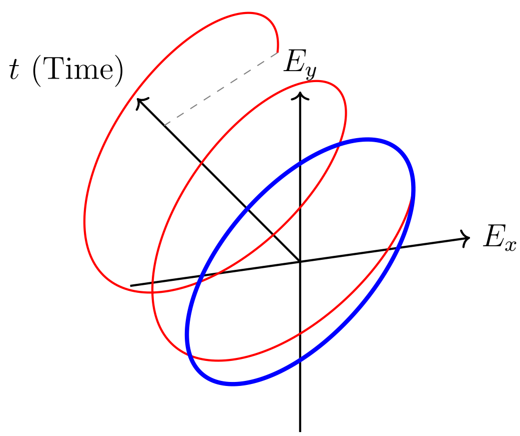
Elliptical ($\Delta\phi = \pi/6$)

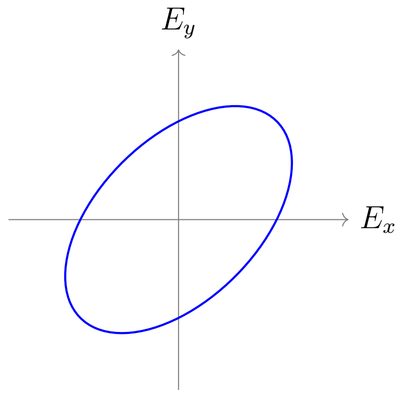
For $\Delta\phi = \frac{\pi}{3}$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 cos(\omega t + \frac{\pi}{3})\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + [E_0 cos(\omega t) cos(\frac{\pi}{3}) - E_0 sin(\omega t) sin(\frac{\pi}{3})]\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + [E_0 cos(\omega t) \cdot \frac{1}{2} - E_0 sin(\omega t) \cdot \frac{\sqrt{3}}{2}]\hat{j}$$



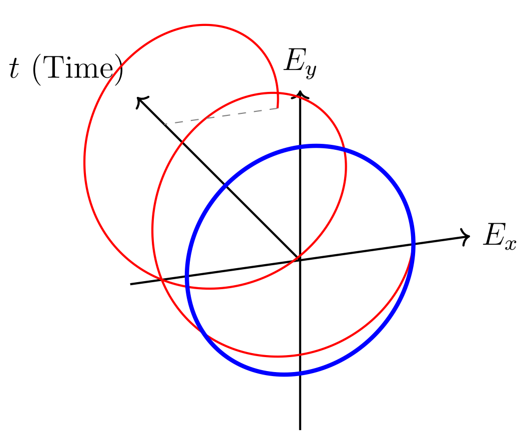


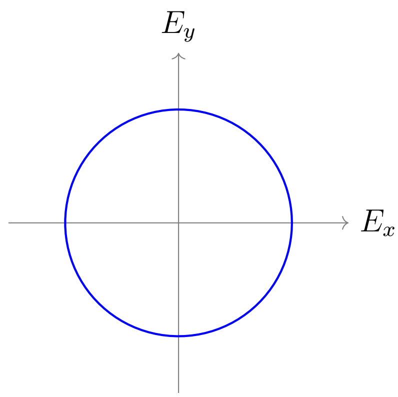
Elliptical ($\Delta\phi = \pi/3$)

For $\Delta\phi = \frac{\pi}{2}$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 cos(\omega t + \frac{\pi}{2})\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 sin(\omega t)\hat{j}$$





Circular ($\Delta\phi = \pi/2$)

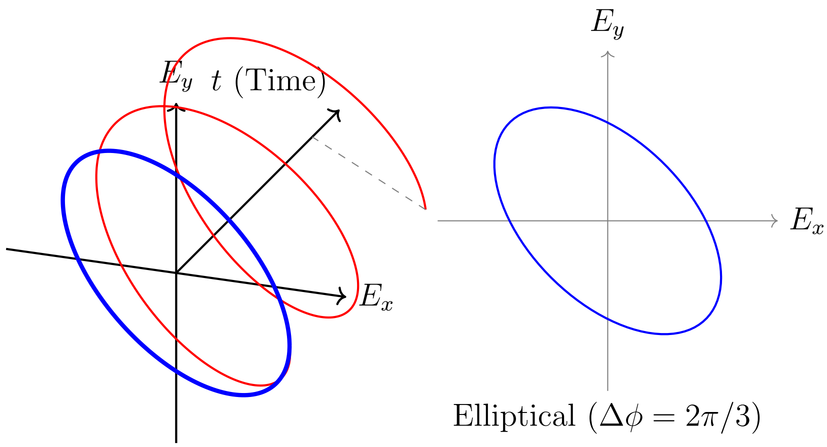
For $\Delta\phi = \frac{2\pi}{3}$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 cos(\omega t + \frac{2\pi}{3})\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + [E_0 cos(\omega t) cos(\frac{2\pi}{3}) - E_0 sin(\omega t) sin(\frac{2\pi}{3})]\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + [E_0 cos(\omega t) \cdot (-\frac{1}{2}) - E_0 sin(\omega t) \cdot \frac{\sqrt{3}}{2}]\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} - [E_0 cos(\omega t) \cdot \frac{1}{2} + E_0 sin(\omega t) \cdot \frac{\sqrt{3}}{2}]\hat{j}$$



Elliptical ($\Delta\phi = 2\pi/3$)

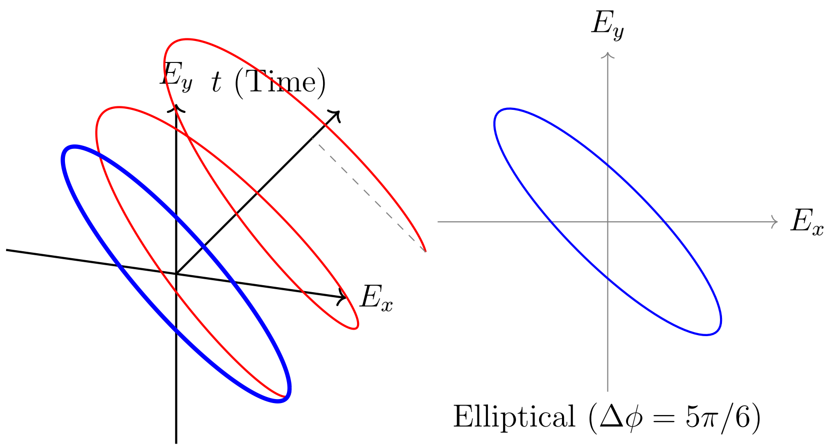
For $\Delta\phi = \frac{5\pi}{6}$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 cos(\omega t + \frac{5\pi}{6})\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + [E_0 cos(\omega t) cos(\frac{5\pi}{6}) - E_0 sin(\omega t) sin(\frac{5\pi}{6})]\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + [E_0 cos(\omega t) \cdot (-\frac{\sqrt{3}}{2}) - E_0 sin(\omega t) \cdot \frac{1}{2}]\hat{j}$$

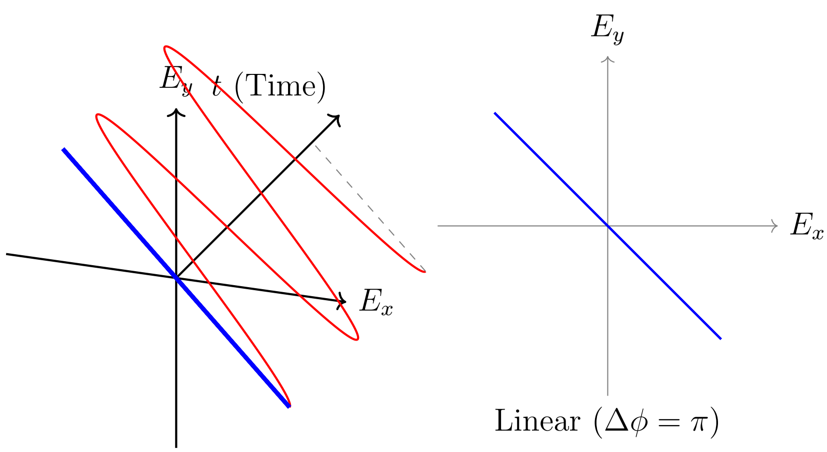
$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} - [E_0 cos(\omega t) \cdot \frac{\sqrt{3}}{2} + E_0 sin(\omega t) \cdot \frac{1}{2}]\hat{j}$$



Elliptical ($\Delta\phi = 5\pi/6$)

For $\Delta\phi = \pi$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + E_0 cos(\omega t + \pi)\hat{j}$$
$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} - E_0 cos(\omega t)\hat{j}$$
$$\vec{E}(t) = E_0 cos(\omega t) \cdot (\hat{i} - \hat{j})$$



Linear ($\Delta\phi = \pi$)

c) Identify which of the above phase differences yield:

- Linear polarization
- Circular polarization
- Elliptical (non-circular) polarization

Explain the physical reason behind each case.

Solution:

- **Linear Polarization:**

$$\Delta\phi = 0, \quad \pi$$

  Physical Reason: The x and y components oscillate either completely in phase ($\Delta\phi = 0$) or completely out of phase ($\Delta\phi = \pi$). In these cases, $\vec{E}(t)$ will be either $E_0 cos(\omega t) \cdot (\hat{\imath} + \hat{\jmath})$ or $E_0 cos(\omega t) \cdot (\hat{\imath} - \hat{\jmath})$, which means they oscillate linearly with the same direction of $\hat{\imath} + \hat{\jmath}$ or $\hat{\imath} - \hat{\jmath}$ respectively.

- **Circular Polarization:**

$$\Delta\phi = \frac{\pi}{2}$$

  Physical Reason: Both x and y components are exactly $\pi/2$ out of phase. Since the amplitudes are equal (E_0), the magnitude of the total electric field vector $|\vec{E}(t)| = \sqrt{E_x^2 + E_y^2} = E_0$ remains constant, while its direction rotates at a constant angular frequency ω.

- **Elliptical Polarization:**

$$\Delta\phi = \frac{\pi}{6}, \ \frac{\pi}{3}, \ \frac{2\pi}{3}, \ \frac{5\pi}{6}$$

  Physical Reason: When the phase difference is arbitrary (neither 0 nor π nor $\pi/2$), although the electric field vector rotates, the magnitude is not constant. Therefore, the polarization will be elliptical

d) Repeat part (b) but with unequal amplitudes:

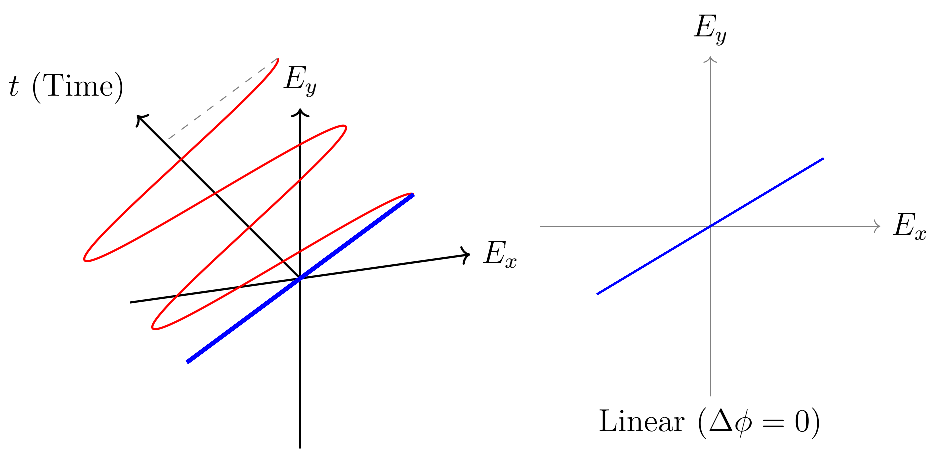
$$E_x(t) = E_0 cos(\omega t) \quad E_y(t) = 0.6E_0 cos(\omega t + \Delta\phi)$$

Discuss how amplitude mismatch affects the shape and orientation of the polarization ellipse.

Solution: I expect that because the max value of the y component of the electromagnetic wave decreases to $0.6E_0$, the heights of the all polarizations will be shorter. However, the widths remain unchanged because the maximum value of the x component remains constant.

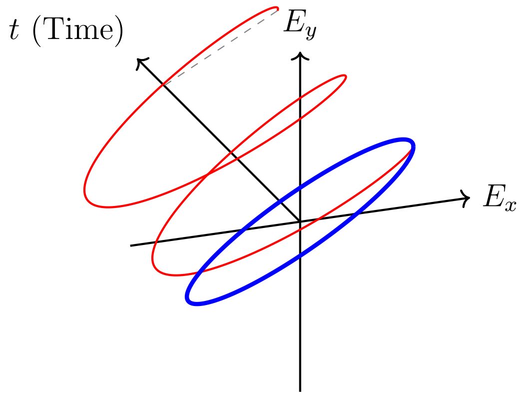
For $\Delta\phi = 0$:

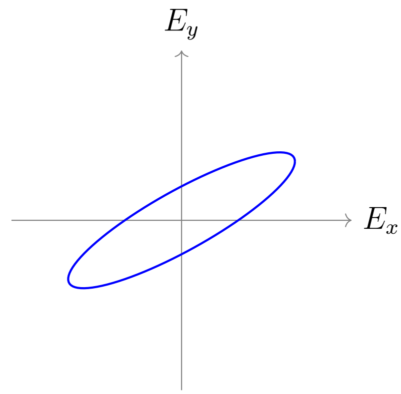
$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6E_0 cos(\omega t)\hat{j}$$
$$\vec{E}(t) = E_0 cos(\omega t) \cdot (\hat{i} + 0.6\hat{j})$$



For $\Delta\phi = \frac{\pi}{6}$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6E_0 cos(\omega t + \frac{\pi}{6})\hat{j}$$
$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6[E_0 cos(\omega t)cos(\frac{\pi}{6}) - E_0 sin(\omega t)sin(\frac{\pi}{6})]\hat{j}$$
$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6[E_0 cos(\omega t) \cdot \frac{\sqrt{3}}{2} - E_0 sin(\omega t) \cdot \frac{1}{2}]\hat{j}$$





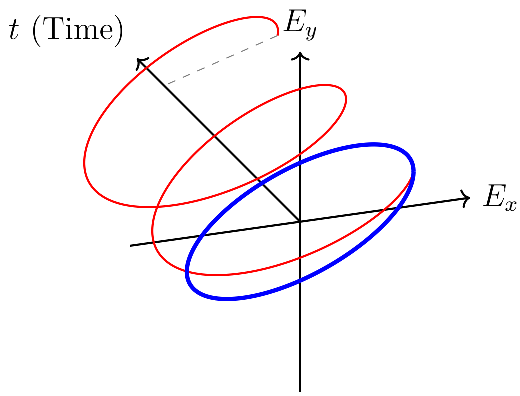
Elliptical ($\Delta\phi = \pi/6$)

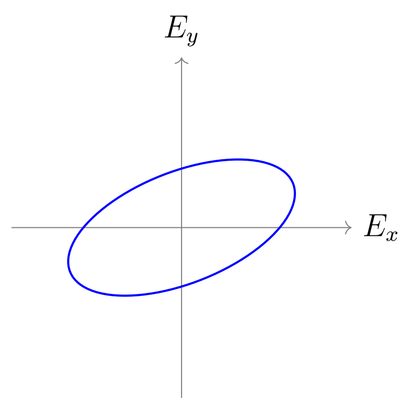
For $\Delta\phi = \frac{\pi}{3}$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6 E_0 cos(\omega t + \frac{\pi}{3})\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6[E_0 cos(\omega t) cos(\frac{\pi}{3}) - E_0 sin(\omega t) sin(\frac{\pi}{3})]\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6[E_0 cos(\omega t) \cdot \frac{1}{2} - E_0 sin(\omega t) \cdot \frac{\sqrt{3}}{2}]\hat{j}$$



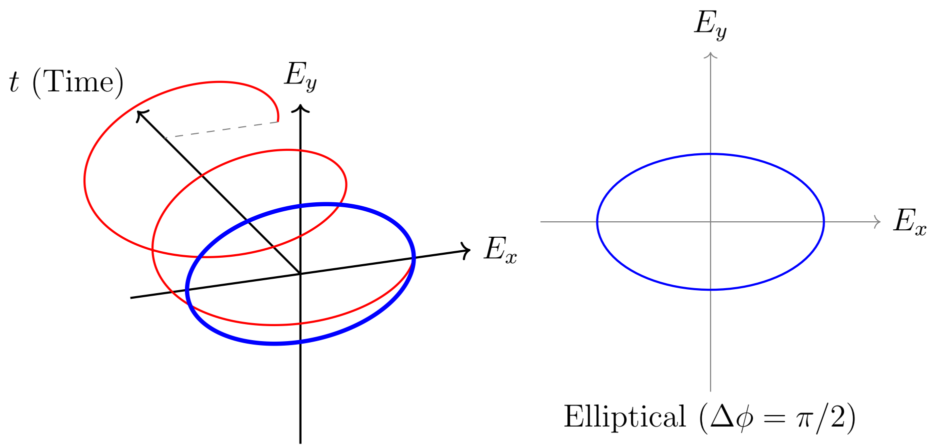


Elliptical ($\Delta\phi = \pi/3$)

For $\Delta\phi = \frac{\pi}{2}$:

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6 E_0 cos(\omega t + \frac{\pi}{2})\hat{j}$$

$$\vec{E}(t) = E_0 cos(\omega t)\hat{i} + 0.6 E_0 sin(\omega t)\hat{j}$$



Elliptical ($\Delta\phi = \pi/2$)

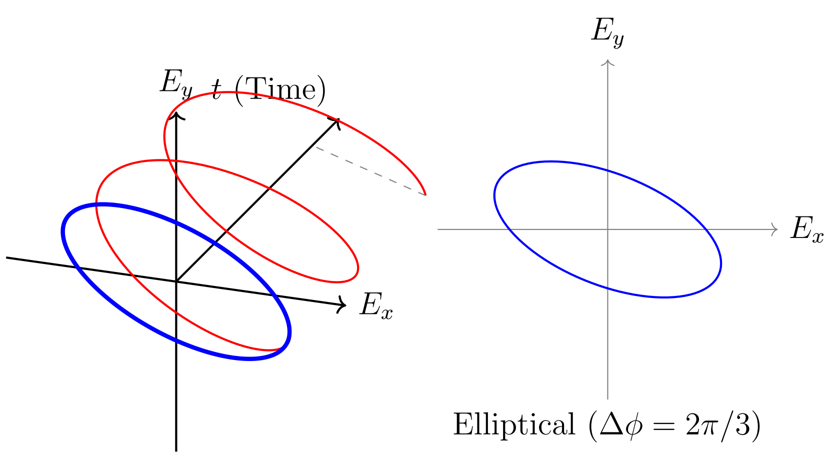
For $\Delta\phi = \frac{2\pi}{3}$:

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} + 0.6E_0cos(\omega t + \frac{2\pi}{3})\hat{j}$$

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} + 0.6[E_0cos(\omega t)cos(\frac{2\pi}{3}) - E_0sin(\omega t)sin(\frac{2\pi}{3})]\hat{j}$$

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} + 0.6[E_0cos(\omega t) \cdot (-\frac{1}{2}) - E_0sin(\omega t) \cdot \frac{\sqrt{3}}{2}]\hat{j}$$

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} - 0.6[E_0cos(\omega t) \cdot \frac{1}{2} + E_0sin(\omega t) \cdot \frac{\sqrt{3}}{2}]\hat{j}$$



Elliptical ($\Delta\phi = 2\pi/3$)

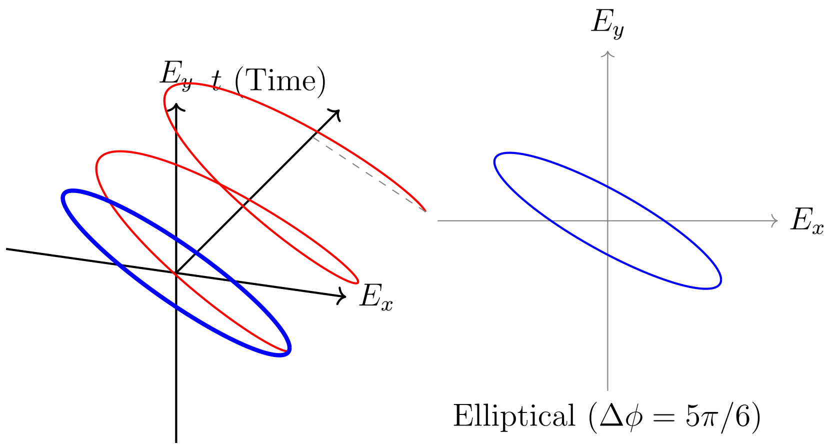
For $\Delta\phi = \frac{5\pi}{6}$:

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} + 0.6E_0cos(\omega t + \frac{5\pi}{6})\hat{j}$$

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} + 0.6[E_0cos(\omega t)cos(\frac{5\pi}{6}) - E_0sin(\omega t)sin(\frac{5\pi}{6})]\hat{j}$$

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} + 0.6[E_0cos(\omega t) \cdot (-\frac{\sqrt{3}}{2}) - E_0sin(\omega t) \cdot \frac{1}{2}]\hat{j}$$

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} - 0.6[E_0cos(\omega t) \cdot \frac{\sqrt{3}}{2} + E_0sin(\omega t) \cdot \frac{1}{2}]\hat{j}$$



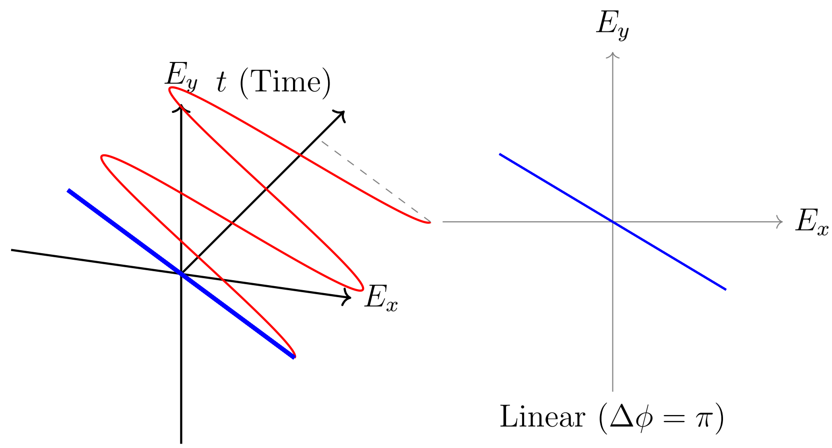
Elliptical ($\Delta\phi = 5\pi/6$)

For $\Delta\phi = \pi$:

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} + 0.6E_0cos(\omega t + \pi)\hat{j}$$

$$\vec{E}(t) = E_0cos(\omega t)\hat{i} - 0.6E_0cos(\omega t)\hat{j}$$

$$\vec{E}(t) = E_0cos(\omega t) \cdot (\hat{i} - 0.6\hat{j})$$



These graphs satisfies my expectations.

e) (d in HW file but it is mistaken) Write a Python or MATLAB code that:

1. Defines $E_x(t)$ and $E_y(t)$,
2. Computes the **instantaneous electric field magnitude**

$$|E(t)| = \sqrt{E_x(t)^2 + E_y(t)^2}$$

3. Plots $|E(t)|$ as a function of time for **different phases given above**.

Solution:

Derivation of the $|E(t)|$ for the general form ($E_x(t) = E_0 cos(\omega t)$, $E_y(t) = E_0 cos(\omega t + \Delta\phi)$):

$$E_x(t) = E_0 cos(\omega t) \quad E_y(t) = E_0 cos(\omega t + \Delta\phi)$$

$$|E(t)| = \sqrt{(E_0 cos(\omega t))^2 + (E_0 cos(\omega t + \Delta\phi))^2}$$

$$|E(t)| = E_0 \sqrt{cos^2(\omega t) + cos^2(\omega t + \Delta\phi)}$$

$$|E(t)| = E_0 \sqrt{\frac{1 + cos(2\omega t)}{2} + \frac{1 + cos(2\omega t + 2\Delta\phi)}{2}}$$

$$|E(t)| = E_0 \sqrt{1 + \frac{cos(2\omega t) + cos(2\omega t + \Delta\phi)}{2}}$$

$$|E(t)| = E_0 \sqrt{1 + cos(2\omega t + \Delta\phi) cos(\Delta\phi)} \tag{1}$$

Equation (1) will be used for most of the calculation in this part.

For $\Delta\phi = 0$:

$$E_x(t) = E_0 cos(\omega t) \qquad E_y(t) = E_0 cos(\omega t)$$
$$|E(t)| = \sqrt{(E_0 cos(\omega t))^2 + (E_0 cos(\omega t))^2}$$
$$|E(t)| = \sqrt{2} E_0 |cos(\omega t)|$$

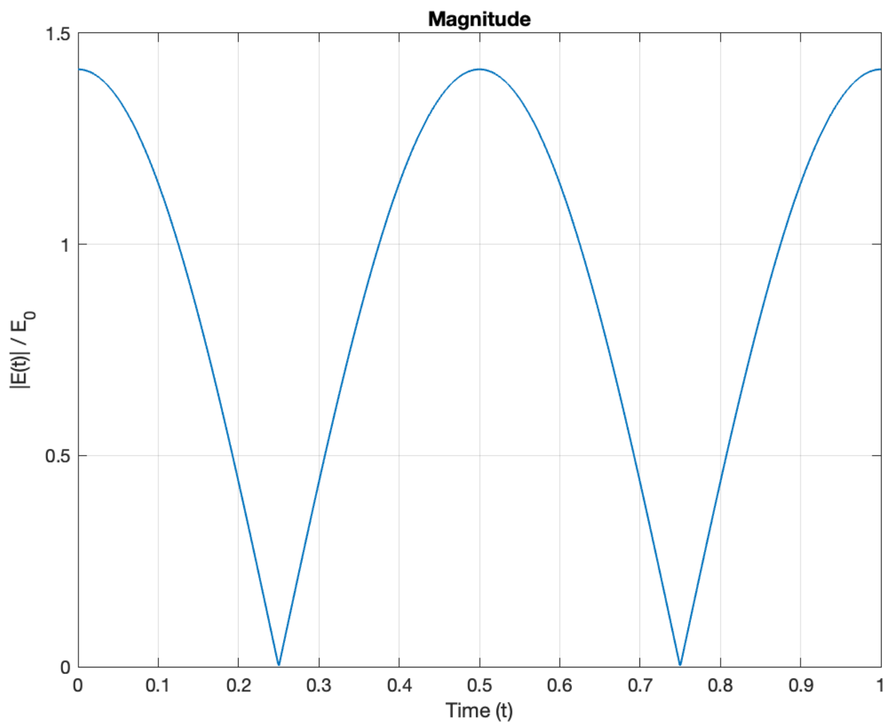


Figure 1: $|E(t)|$ for $\Delta\phi = 0$

For $\Delta\phi = \frac{\pi}{6}$:

$$E_x(t) = E_0 cos(\omega t) \qquad E_y(t) = E_0 cos(\omega t + \frac{\pi}{6})$$

$$|E(t)| = E_0 \sqrt{1 + cos(2\omega t + \frac{\pi}{6}) cos(\frac{\pi}{6})}$$

$$|E(t)| = E_0 \sqrt{1 + \frac{\sqrt{3}}{2} cos(2\omega t + \frac{\pi}{6})}$$

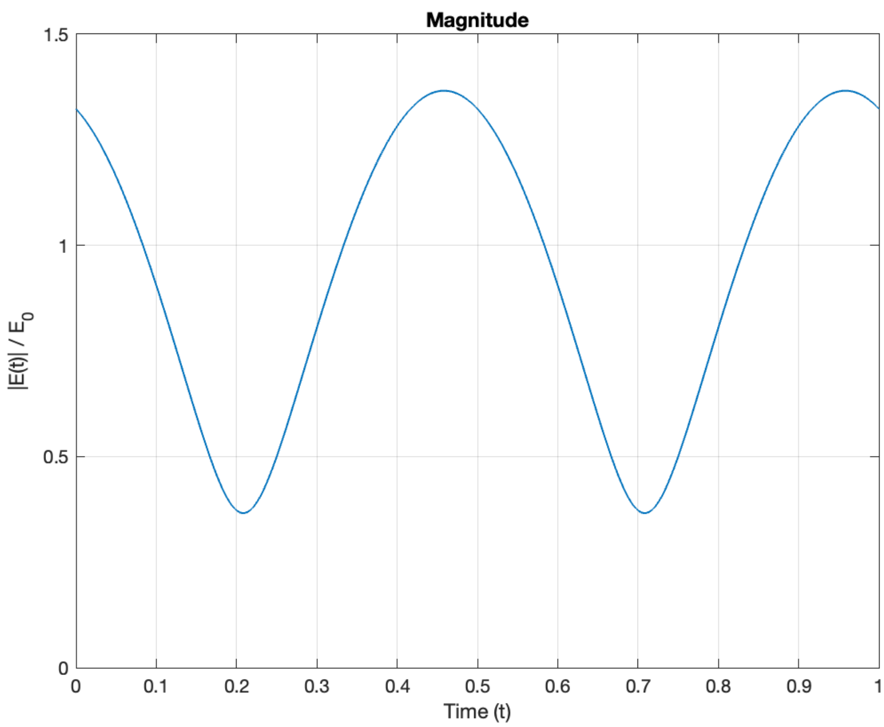


Figure 2: $|E(t)|$ for $\Delta\phi = \pi/6$

For $\Delta\phi = \frac{\pi}{3}$:

$$E_x(t) = E_0 cos(\omega t) \qquad E_y(t) = E_0 cos(\omega t + \frac{\pi}{3})$$

$$|E(t)| = E_0 \sqrt{1 + cos(2\omega t + \frac{\pi}{3}) cos(\frac{\pi}{3})}$$

$$|E(t)| = E_0 \sqrt{1 + \frac{1}{2} cos(2\omega t + \frac{\pi}{3})}$$

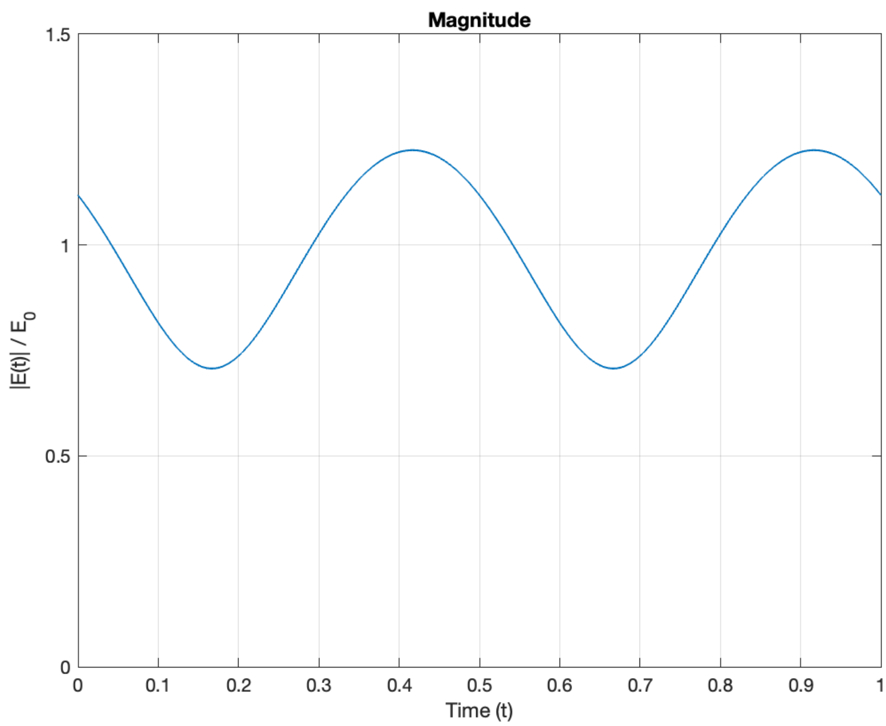


Figure 3: $|E(t)|$ for $\Delta\phi = \pi/3$

For $\Delta\phi = \frac{\pi}{2}$:

$$E_x(t) = E_0 cos(\omega t) \qquad E_y(t) = E_0 cos(\omega t + \frac{\pi}{2}) = E_0 sin(\omega t)$$

$$|E(t)| = E_0 \sqrt{cos^2(\omega t) + sin^2(\omega t)} = E_0$$

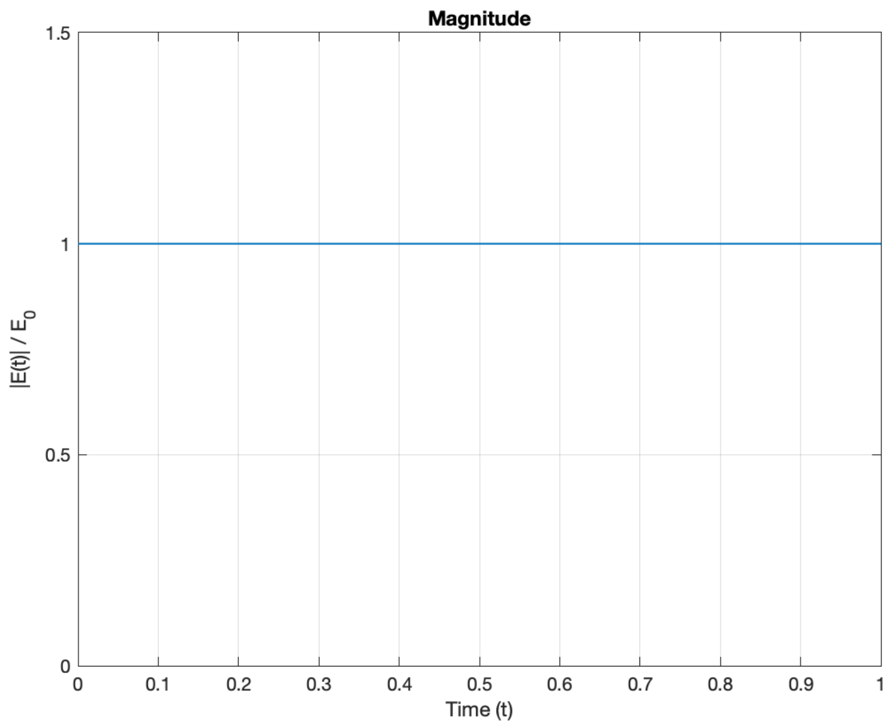


Figure 4: $|E(t)|$ for $\Delta\phi = \pi/2$

For $\Delta\phi = \frac{2\pi}{3}$:

$$E_x(t) = E_0 cos(\omega t) \qquad E_y(t) = E_0 cos(\omega t + \frac{2\pi}{3})$$

$$|E(t)| = E_0\sqrt{1 + cos(2\omega t + \frac{2\pi}{3})cos(\frac{2\pi}{3})}$$

$$|E(t)| = E_0\sqrt{1 - \frac{1}{2}cos(2\omega t + \frac{2\pi}{3})}$$

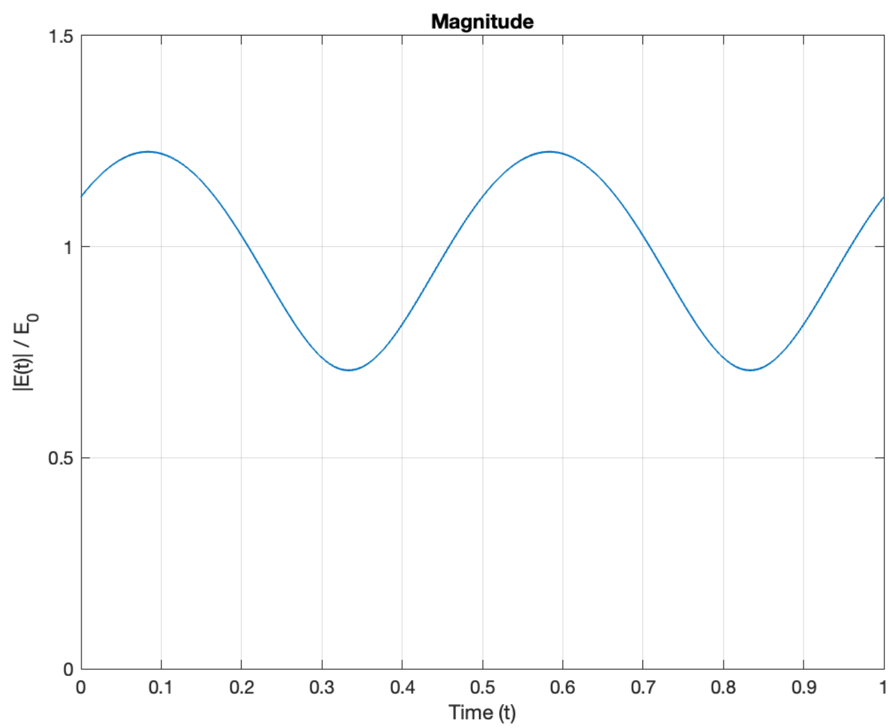


Figure 5: $|E(t)|$ for $\Delta\phi = 2\pi/3$

For $\Delta\phi = \frac{5\pi}{6}$:

$$E_x(t) = E_0 cos(\omega t) \qquad E_y(t) = E_0 cos(\omega t + \frac{5\pi}{6})$$
$$|E(t)| = E_0\sqrt{1 + cos(2\omega t + \frac{5\pi}{6})cos(\frac{5\pi}{6})}$$
$$|E(t)| = E_0\sqrt{1 - \frac{\sqrt{3}}{2}cos(2\omega t + \frac{5\pi}{6})}$$

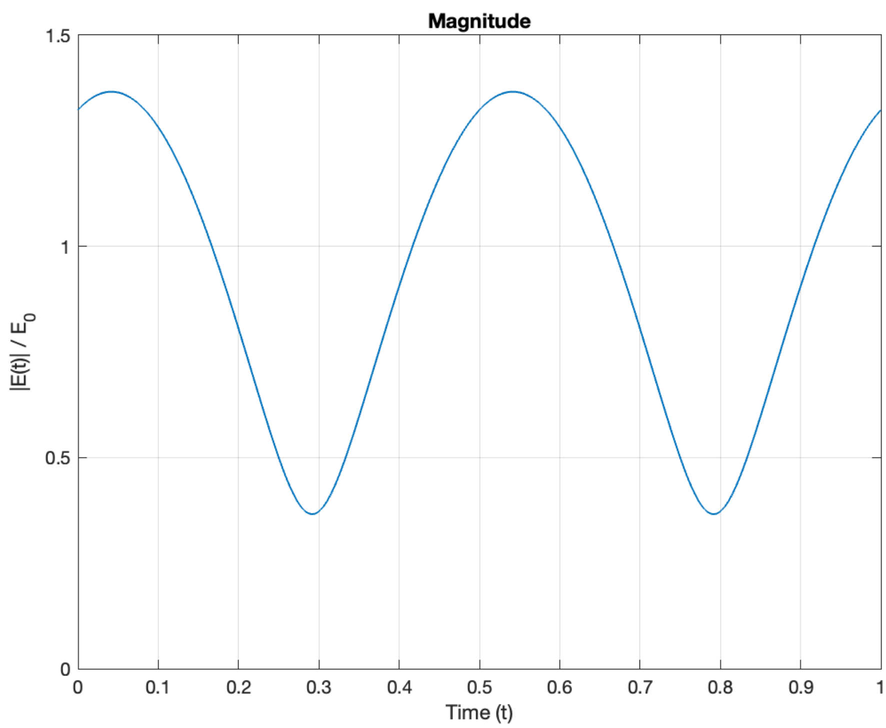


Figure 6: $|E(t)|$ for $\Delta\phi = 5\pi/6$

For $\Delta\phi = \pi$:

$$E_x(t) = E_0 cos(\omega t) \qquad E_y(t) = E_0 cos(\omega t + \pi) = -E_0 cos(\omega t)$$
$$|E(t)| = \sqrt{(E_0 cos(\omega t))^2 + (-E_0 cos(\omega t))^2}$$
$$|E(t)| = \sqrt{2} E_0 |cos(\omega t)|$$

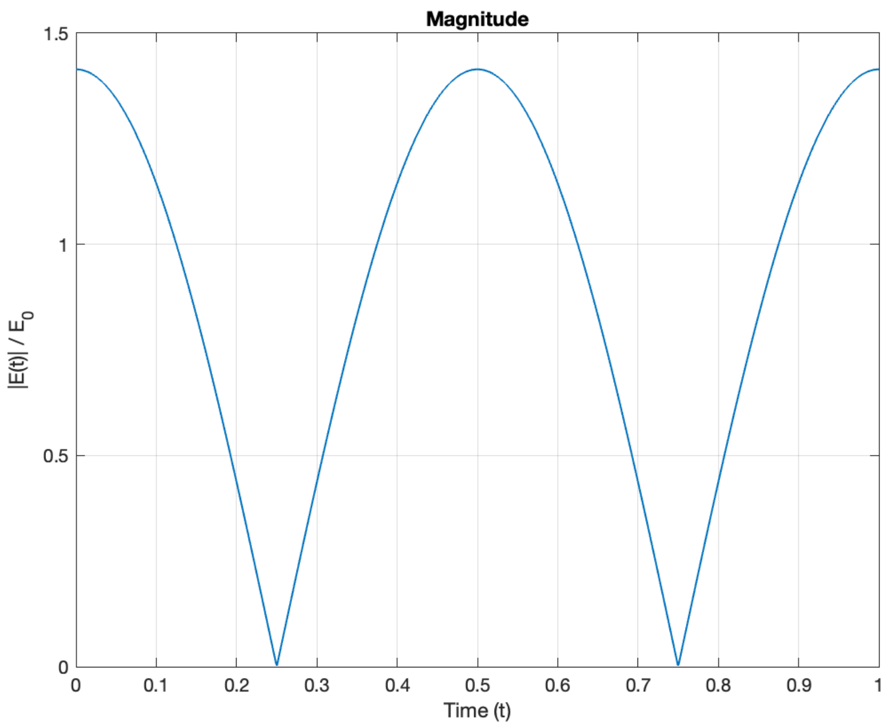


Figure 7: $|E(t)|$ for $\Delta\phi = \pi$

```matlab
E0 = 1;
f = 1;
w = 2*pi*f;
t = linspace(0, 1, 1000);

x_values = 0:6;
file_names = {'0', 'pi_6', 'pi_3', 'pi_2', '2pi_3', '5pi_6', 'pi'};

for k = 1:length(x_values)
    x = x_values(k);
    phase = x * pi/6;

    Ex = E0 * cos(w*t);
    Ey = E0 * cos(w*t + phase);
    E_mag = sqrt(Ex.^2 + Ey.^2);

    fig = figure('Visible', 'off');
    plot(t, E_mag, 'LineWidth', 1);
    hold on;
    grid on;

    title('Magnitude');
    xlabel('Time (t)');
    ylabel('|E(t)| / E_0');
    ylim([0 1.5]);

    filename = ['hw5_', file_names{k}, '_plot.png'];
    saveas(fig, filename);
    close(fig);
end
```
Listing 1: MATLAB implementation

Question 2

Light is incident from **air** ($n_1 = 1.0$) onto a **plastic surface** with refractive index $n_2 = 1.5$. When unpolarized light reflects from a dielectric interface, the reflected and transmitted (refracted) rays become orthogonal at the **Brewster angle**, and the reflected light becomes perfectly **linearly polarized**.

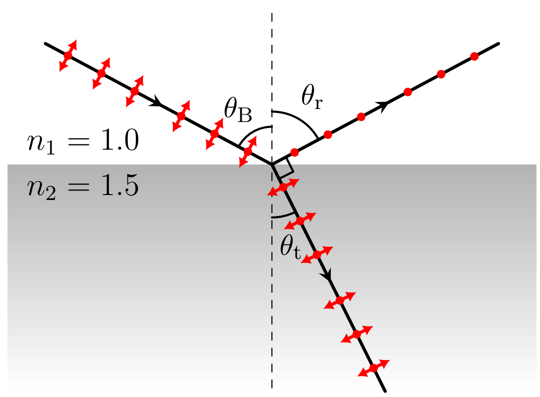
a) Using Snell's law, derive the condition for Brewster's angle θ_B, defined by the requirement that the reflected and transmitted rays are at 90° to each other:

$$\theta_r + \theta_t = 90°$$

Show that:

$$tan\theta_B = \frac{n_2}{n_1}$$

Solution:



$$\theta_B = \theta_r$$

So,

$$\theta_t = 90° - \theta_B$$

By Snell's law:

$$n_1 sin\theta_B = n_2 sin\theta_t$$

$$n_1 sin\theta_B = n_2 sin(90° - \theta_B)$$

$$\frac{n_2}{n_1} = \frac{sin\theta_B}{sin(90^\circ - \theta_B)} = \frac{sin\theta_B}{cos\theta_B}$$

Answer

$$\frac{n_2}{n_1} = tan\theta_B$$

b) Calculate the **Brewster angle** for air-plastic interface:

$$n_1 = 1.0 \quad n_2 = 1.5$$

Provide your answer in degrees.

Solution:

By the part a, we know that:

$$\frac{n_2}{n_1} = tan\theta_B$$

so,

$$\theta_B = arctan(\frac{n_2}{n_1})$$

Answer

$$\theta_B \approx 56^\circ$$

c) For this angle, determine whether the reflected light is **s-polarized** or **p-polarized**, and briefly explain why.

Solution: The incident lay is an electromagnetic wave, so it affect the electrons to oscillate in glass because of the electric field. Both components of the polarization cause this oscillation and their directions are the same as the oscillation direction of the electric field of the refraction lay. This oscillation radiates the reflection. But, at Brewster's angle, the reflection and refraction are orthogonal to each other, so p-polarization component cannot reflect because the oscillation of the electron which causes this polarization is at the same direction as the reflected lay. Therefore, only s-polarized lay reflects.

Answer

s-polarized

Question 3

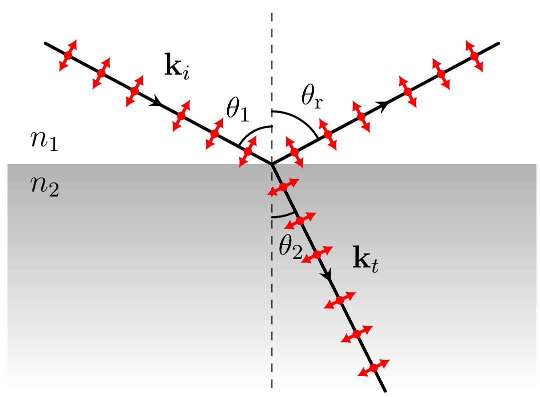
Consider a monochromatic light ray incident from **air** ($n_1 = 1.0$) onto a **glass** medium (n_2) at a flat interface (the boundary lies in the x-y plane, normal along z). The angle of incidence is θ_1(in air) and the angle of refraction is θ_2(in glass). Focus on the **component of the wavevector parallel to the interface** (take this as the x-direction). Using the fact that the **parallel component of momentum (or wavevector) must be continuous** across the interface,

$$k_{i,x} = k_{t,x}$$

Show that this condition leads to **Snell's law**:

$$n_1 sin\theta_1 = n_2 sin\theta_2$$

Solution:



k_i and k_t are the magnitudes of the wavevectors $\mathbf{k}_i$ and $\mathbf{k}_t$, respectively.

$$k_i = n_1 \frac{\omega}{c} \qquad k_t = n_2 \frac{\omega}{c}$$

$$k_{i,x} = k_i \cdot sin\theta_1 \qquad k_{t,x} = k_t \cdot sin\theta_2$$

$$k_{i,x} = n_1 \frac{\omega}{c} \cdot sin\theta_1 \qquad k_{t,x} = n_2 \frac{\omega}{c} \cdot sin\theta_2$$

By the fact in the question,

$$n_1 \frac{\omega}{c} \cdot sin\theta_1 = n_2 \frac{\omega}{c} \cdot sin\theta_2$$

imply that with cancel ω/c

Answer

$$n_1 sin\theta_1 = n_2 sin\theta_2$$

Question 4

A laser with power $P = 100$ mW and wavelength $\lambda = 532$ nm is focused to a circular spot of radius $r = 0.5$ mm on a **perfectly reflecting mirror**.

1. Calculate the **intensity** I.

Solution:

$$I = \frac{P}{A}$$

For our question:

$$I = \frac{100\text{mW}}{\pi(0.5\text{mm})^2} = \frac{100 \cdot 10^{-3}\text{W}}{\pi(0.5 \cdot 10^{-3}\text{m})^2}$$

Answer

$$I = \frac{4}{\pi} \cdot 10^5 \ \text{Wm}^{-2}$$

2. Calculate the **radiation pressure** P_{rad}.

Solution:

Let **p** refers to the momentum of the light, and p is the magnitude of **p** and can be calculated with the given equation. ($c = 3.00 \times 10^8$ m/s is the speed of light, U=$P \cdot \Delta t$ is energy)

$$p = \frac{U}{c} = \frac{P \cdot \Delta t}{c} = \frac{I \cdot A \cdot \Delta t}{c}$$

Because the mirror is perfectly reflecting, our incident ray will be reflected with the reflection angle 0. Therefore,

$$|\Delta \mathbf{p}| = 2p = 2\frac{I \cdot A \cdot \Delta t}{c}$$

This $\Delta \mathbf{p}$ is caused by the interaction between mirror and the electromagnetic wave, so the wave apply a force F and it causes a pressure which is P_{rad}. Now, let us calculate them.

$$F = \frac{\Delta \mathbf{p}}{\Delta t} = \frac{2\frac{I \cdot A \cdot \Delta t}{c}}{\Delta t} = 2\frac{I \cdot A}{c}$$

$$P_{rad} = \frac{F}{A} = \frac{2\frac{I \cdot A}{c}}{A} = 2\frac{I}{c}$$

For our question:

$$P_{rad} = 2\frac{\frac{4}{\pi} \cdot 10^5 \text{ Wm}^{-2}}{3 \cdot 10^8 \text{ m/s}}$$

Answer

$$P_{rad} = \frac{8}{3\pi} \cdot 10^{-3} \text{ Nm}^{-2}$$

3. Find the **force** exerted on the mirror and perfectly absorbing surfaces.

Solution:

As I mentioned in the previous part,

$$F = 2\frac{I \cdot A}{c} = P_{rad} \cdot A$$

With replacing the variables with the corresponding values,

$$F = \frac{8}{3\pi} \cdot 10^{-3} \text{ Nm}^{-2} \cdot \pi \cdot 25 \cdot 10^{-8} \text{ m}^2$$

Answer

$$F = \frac{2}{3} \cdot 10^{-9} \text{ N}$$